Microeconomic Foundations, I:
Choice and Competitive Markets

Instructor's Manual

Chapter 3: Basics of Consumer Demand

Notes on the Chapter

This chapter introduces the consumer's problem, and then

1. provides basic results: existence and homogeneity of solutions, convexity of the solution set, exhaustion of wealth under local insatiability,
2. applies Berge's Theorem to the problem, deriving upper semi-continuity of Marshallian demand and continuity of indirect utility, and then
3. discusses solution of the problem using calculus, emphasizing the intuition of the "equal bang-for-the-buck" conditions.

Most students will find the chapter relatively straightforward *if* they are comfortable with the mathematics surrounding Berge's Theorem, which is to say: For most students, the material surrounding Berge's Theorem (in Appendix 4) will be where difficulties are encountered, if any are. The basics of constrained optimization, related in Appendix 5, may also be largely new.

So, despite the relative simplicity of the material in this chapter (at least, relative to Chapters 1 and 2) you may want spent time in class going through material from Appendix 4 and, depending on the level of preparation of your students, Appendix 5. Looking ahead a bit, Chapter 4 (on Afriat's Theorem) provides very little to discuss in class, unless you plan to go through the proof (and good luck to that!). So, with two 95-minute class sessions a week, I've been able to cover Chapters 3 and 4 in two sessions, with Chapter 3 and the accompanying appendices taking a session and a half and Chapter 4 the last half-session.

With regard to Appendix 4, I call to your attention the two simple exercises in the Appendix. I provide solutions here, following solutions to the unstarred problems. If you assign this Appendix formally (as opposed to telling students "It is there if you need it"), you should consider making the final section optional; it ties together the concepts of upper and lower semi-continuity of functions and of correspondences, but it it is likely to frustrate readers who are not mathematically sophisticated.

Finally, I call your attention to Problem 3.14 in the text. This problem gives students the opportunity to apply Berge's Theorem to a relatively simple setting, and also to work a bit on identifying assumptions that guarantee quasi-concavity of an objective function. Also (in case you need more inducement to assign and discuss this problem), this problem provides an excellent example with which to introduce comparative statics, both traditional (using the Implicit Function Theorem) and using monotone methods. After much to-ing and fro-ing, I decided not to include a chapter on comparative statics. But if you decide to include this material, asking them to give conditions under which Mr. Crusoe's first- and second-period levels of consumption are increasing in his endowment of seed corn is a great entry into a discussion of traditional and monotone c.s.

Notes on the Problems

I've already discussed Problem 3.14. The problems are otherwise fairly straightforward. For a mathematically adept group, I recommend 3.11.

Solutions to Non-Starred Problems

■ 3.2. If prices are strictly positive, then the consumer chooses to consume 0 units of goods with index larger than j. To see this, fix the consumer's income y and let x^* be any bundle such that $x^*_k > 0$ for at least one index $k > j$ and $p \cdot x^* \le y$. Let x' be the bundle given by $x'_i = x^*_i$ for $i = 1, \dots j$ and $x'_i = 0$ for $i \ge j$. Then, per the problem's assumptions, $x^* \sim x'$. Moreover, $p \cdot x' < p \cdot x^*$, since x' costs the same as x^* for commodities 1 through j, costs no more for $j+1$ through k, and costs strictly less for those indices $k > j$ with $x^*_k > 0$. But then, since $p \cdot x^* \le y$, where y is the consumer's (given) income, $p \cdot x' < y$, and by local insatiability, there is some bundle x'' such that $x'' \succ x' \sim x^*$ and $p \cdot x'' \le y$. Thus x^* cannot be optimal.

Now suppose that, for given y and p, there are distinct solutions to the CP, call them x^1 and x^2. By the previous paragraph, we know that $x^1_k = x^2_k = 0$ for all indices $k > j$. Hence the two must differ for some commodity index $k \le j$. But then, because preferences are strictly convex in the first j goods, the bundle $x' = 0.5x^1 + 0.5x^2 \succ x^1 \sim x^2$. Of course this convex combination of x^1 and x^2 is affordable at prices p with income y (the budget set is convex), contradicting the supposed optimality of x^1 and x^2.

■ 3.3. (a, b) This utility function is strictly concave, so there is a unique solution to the CP. Moreover, at this solution, the demand for good i depends only on the price of good i, p_i, and the income level y: $x_i(p, y) = (\beta_i y)/(p_i \sum_j \beta_j)$. With the substitution of αs for βs, this is identical to demand with Cobb-Douglas utility, for the simple reason that this utility function is a (strict) monotone transformation of Cobb-Douglas utility, as specified beginning on page 67. (Define $v(s) = \ln(u(x))$ for u given in the first display on page 67, noting that this transforms utility level 0 into $-\infty$.)

(c) In place of footnote 5, one could write "Or apply the strictly monotone transformation $v(x) = \ln(u(x))$ to get the utility function $v(x) = \sum_i \alpha_i \ln(x_i)$, which is strictly concave." Of course, this utility function does take on the value $-\infty$ whenever one or more components of x are zero, and some special pleading to get the results of Appendix 5 to apply are needed.

■ 3.5. Apply the general method developed in the solution to Problem 3.8 (solution given in the *Student's Guide*.

■ 3.7. Here is a sketch: First, we can drop the "outside" exponent $1/\mu$ from the utility function, since doing so is a strictly increasing transformation of the original utility function. Then the bang-for-the-buck of good j is

$$\frac{\alpha_j \mu x_j^{\mu-1}}{p_j} \; ,$$

which approaches infinity as x_j approaches zero. (Remember that $0 < \mu < 1$.) Therefore, for the first-order/complementary-slackness conditions to hold, all the x_j must be strictly positive. And then the first-order/complementary-slackness conditions are that the bangs-for-the-bucks must all be equal to one another. Letting λ be the common value of the BFBs, we have

$$\lambda = \frac{\alpha_j \mu x_j^{\mu-1}}{p_j} \text{ for all } j\text{, which implies that } x_j = \left(\frac{p_j \lambda}{\alpha_j \mu}\right)^{1/(\mu-1)}.$$

We know the optimal values of the x_j if we know λ, but λ is easily found by computing

$$p_j x_j = p_j \left(\frac{p_j \lambda}{\alpha_j \mu}\right)^{1/(\mu-1)} \text{ hence } \sum_{j=1}^{k} p_j \left(\frac{p_j \lambda}{\alpha_j \mu}\right)^{1/(\mu-1)} = y.$$

I leave to you the straightforward algebra of pulling λ out of the summation and isolating it in an equation.

The case $\mu = 1$ is $u(x) = \sum_j \alpha_j x_j$, or linear indifference curves. See Problem 3.6(a). And as μ approaches zero, we see from the second display above that x_j approaches a constant times α_j/p_j. That is, expenditure on good j approaches a constant fraction $\alpha_j/\sum_{j'} \alpha_{j'}$ of total income, which we know from the example worked out in the text is just Cobb-Douglas demand.

■ 3.10. (a) The preferences in 3.3a, 3.6a, 3.6b, and 3.7 are homothetic.

(b) Suppose a consumer has homothetic preferences. If she demands x at prices p and income y, for $y > 0$, then her demand at prices p and incomes y' is (more properly, includes) $(y'/y)x$.

To prove this, we first dispose of the case $y' = 0$: If $y' = 0$, then the only bundle she can afford at any (strictly positive) prices is 0, so this must be her demand. But $(y'/y)x = 0$ if $y' = 0$, so the statement is proven true.

So we can suppose that $y' > 0$. Suppose some budget feasible x' is strictly better than $(y'/y)x$ at p and y'. Then by homotheticity, $(y/y')x'$ is strictly better than $(y/y')[(y'/y)x] = x$. But since $p \cdot x' \leq y'$, $p \cdot [(y/y')x'] = (y/y')[p \cdot x'] \leq (y/y')y' = y$; that is, $(y/y')x$ is affordable at p and y, contradicting the supposed optimality of x at p and y.

▪ 3.11. (a) Proposition 3.5 concerns the case where $y > 0$, so I assume this is so. (And, in any case, since this problem concerns sufficiency of the optimality conditions, and since those conditions involve feasibility, if $y = 0$, the only bundle that could satisfy the conditions is $x = 0$, which is optimal.)

The optimality conditions are that the proposed solution is feasible and that

$$\left.\frac{\partial u}{\partial x_i}\right|_{x^*} \leq \lambda p_i, \text{ for all } i,$$

with strict equality if $x_i > 0$, and with $\lambda = 0$ if $p \cdot x^* < y$.

I assert that $\lambda > 0$. To see why, note that $\partial u/\partial x_1|_{x^*} \leq \lambda p_1$, but by assumption, $\partial u/\partial x_1 > 0$ everywhere. And since $\lambda > 0$, $p \cdot x^* = y$.

Now suppose that x^* is not optimal: $u(\hat{x}) > u(x^*)$ for some $\hat{x}$ such that $p \cdot \hat{x} \leq y$. We can assume w.l.o.g. that $\hat{x} \neq 0$: Since $y > 0$ and $\partial u/\partial x_1 > 0$, if $\hat{x} = 0$, there would be a better (affordable) bundle consisting of a bit of x_1, and then we could replace $\hat{x}$ by this bundle and proceed.

But if $\hat{x} \neq 0$ and $u(\hat{x}) > u(x^*)$, we can find a bundle $\check{x}$ such that $u(\check{x}) > u(x^*)$ and $p \cdot \check{x} < y$. (if $\hat{x}_i > 0$, decrease the amount of commodity i a bit to form $\check{x}$, relying on continuity of u to guarantee that $u(\check{x}) > u(x^*)$.

By the quasi-concavity of u, $u(x^* + a(\check{x} - x^*)) \geq u(x^*)$ for all $a \in (0, 1)$, and in particular for a near zero. But by Taylor's Theorem, for small a,

$$u(x^* + a(\check{x} - x^*)) = u(x^*) + \sum_{i=1}^{k} \left.\frac{\partial u}{\partial x_i}\right|_{x^*} a(\check{x}_i - x_i^*) + o(a).$$

If $x^* > 0$, then $\partial u/\partial x_i = \lambda p_i$ and so $(\partial u/\partial x_i)a(\check{x}_i - x_i^*) = \lambda p_i a(\check{x}_i - x_i^*)$, while if $x^* = 0$, $\partial u/\partial x_i \leq \lambda p_i$ and therefore $(\partial u/\partial x_i)a(\check{x}_i - x_i^*) = (\partial u/\partial x_i)a\check{x}_i \leq \lambda p_i a\check{x}_i = \lambda p_i a(\check{x}_i - x_i^*)$; summing these equations and inequalities gives

$$u(x^* + a(\check{x} - x^*)) \leq u(x^*) + a\lambda \sum_{i=1}^{k} p_i(\check{x}_i - x_i^*) + o(a).$$

But by construction, $p \cdot \check{x} < y = p \cdot x^*$, so $\lambda \sum_{i=1}^{k} p_i(\check{x}_i - x_i^*) < 0$. For small enough a, this says that $u(x^* + a(\check{x} - x^*)) < u(x^*)$, a contradiction.

(b) Suppose $k = 1$ and $u(x) = (x-3)^3$. Let $p_1 = 1$ and $y = 10$. Since u is strictly increasing and a function of one variable, it is manifestly quasi-concave. The solution for the consumers problem is $x = 10$, of course. But $x = 3, \lambda = 0$ satisfies the optimality conditions. Of course, this happens because at $x = 3$, $du/dx = 0$.

■ 3.13. Let u_i^+ be the right-hand partial derivative of u with respect to x_i (that is, for marginal increases in x_i) and let u_i^- be the left-hand partial. Suppose x^* is an alleged solution to the CP. If the budget constraint does not bind at x^*, then the value of x_i for each i can be increased on the margin, so $u_i^+ \le 0$ is necessary. If $x_i^* > 0$, then the x_i can be decreased on the margin, so $u_i^- \ge 0$ is necessary. (If $x_i^* = 0$, then x_i cannot be decreased on the margin, but also it isn't clear what u_i^- is, so we needn't worry about it.)

So, to summarize so far: If x^* is such that $p \cdot x^* < y$, then $u_i^+(x^*) \le 0$ is necessary for all i, and $u_i^-(x^*) \ge 0$ for all i such that $x_i^* > 0$ is necessary.

In the case $p \cdot x^* = y$: We still require that $u_i^-(x^*) \ge 0$ for all i such that $x_i^* > 0$. And, in addition, $u_j^+(x^*)/p_j - u_i^-(x^*)/p_i \le 0$ for all j and i such that $x_i^* > 0$: This arises from a marginal shift that increases x_j at the expense of lowering x_i, in a ratio that leaves you budget feasible.

■ 3.14. To get existence of a solution, we need to be maximizing a continuous function on a compact set. The function being maximized is

$$x \to U(e - x, f(x)),$$

with $x \in [0, e]$, so compactness of the feasible set is automatic. To get that this function is continuous (in x), assume that both f and U are continuous and then apply the following "math facts" about continuous functions: The function $e - x$ is continuous in x; if g and h are continuous functions, then so is the vector function (g, h); and the composition of two continuous functions is continuous.

For convexity of the solution set, we need to know that the function $x \to U(e - x, f(x))$ is quasi-concave. Suppose U is quasi-concave and nondecreasing in its second argument, and f is nondecreasing and concave. (Note, f has to be concave, at least for my proof.) Then if we take x, x', and $a \in [0, 1]$,

$$\begin{aligned}
&U(e - (ax + (1-a)x'), f(ax + (1-a)x')) \\
&\qquad \ge U(e - (ax + (1-a)x'), af(x) + (1-a)f(x')) \\
&\qquad = U(a(e - x, f(x)) + (1-a)(e - x', f(x')) \\
&\qquad \ge \min\{U(e - x, f(x)), U(e - x', f(x'))\}.
\end{aligned}$$

In this sequence, the first inequality uses the concavity of f to conclude that $f(ax+(1-a)x') \geq af(x)+(1-a)f(x')$ and the nondecreasing-in-its-second-argument property of U to conclude that this will not lower the value of U; the equality is simply a matter of using vector notation; and the last inequality is quasi-concavity of U.

For uniqueness of the solution, we need that the basic function $x \to U(e-x, f(x))$ is strictly quasi-concave. Take $x \neq x'$ and $a \in (0,1)$. Make all the assumptions made before and add that U is strictly quasi-concave. Then we are there, since $x \neq x'$ implies that $(e-x, f(x)) \neq (e-x', f(x'))$ by virtue of the first component. Alternatively, assume that U is strictly increasing in its second argument and that f is strictly concave; then the first inequality in the chain becomes strict, since $f(ax+(1-a)x') > af(x)+(1-a)f(x')$, and the strictly-increasing-in-its-second-argument property kicks in.

The final part of this problem entails an application of Berge's Theorem. The constraint correspondence $e \Rightarrow [0,e]$ is clearly compact valued, locally bounded, and both upper and lower semi-continuous. So as long as we know that $(e,x) \to U(e-x, f(x))$ is continuous in (e,x), we can apply Berge's Theorem to see that X^* is upper semi-continuous and v is continuous. To get continuity of the objective function, assume that U is continuous in both arguments and f is continuous, and continuity follows from standard arguments.

As for lower semi-continuity of X^*, you could make enough assumptions so that X^* is singleton valued, hence (automatically) given by a continuous function, hence a continuous correspondence. We already discussed what it takes to ensure that X^* is singleton.

■ Exercise 1 from Appendix 4. Let $X = R$ and

$$f(x) = \begin{cases} \{1/x\}, & \text{for } x \neq 0, \text{ and} \\ \{0\}, & \text{for } x = 0. \end{cases}$$

This satisfies all the requirements: It is singleton valued, its graph is a closed set, and it is not locally bounded at $x = 0$.

■ Exercise 2 from Appendix 4. The graphs of the two correspondences are shown in panels a and b of Figure M4.1 Panel a depicts ϕ_1; the dashed line at $x = 2$ indicates that the parallelogram belonging to the graph of ϕ_1 is open there. Panel b depicts ϕ_2.

It is obvious from the figures that, as the graph in panel b is closed and the graph in panel a is not, ϕ_1 is not upper semi-continuous, but ϕ_2 is. To render ϕ_1 upper semi-continuous, you must take the closure of its graph, which means enlarging $\phi_1(2)$ to be the interval $[0,2]$.

As for lower semi-continuity, neither is. The correspondence ϕ_1 fails at $x = 3$, because range values other than 0 cannot be approached in $\phi_1(x)$ for x approaching 3 from above. If we redefine $\phi_1(3) = \{0\}$, then the correspondence becomes lower semi-continuous (although this takes an argument). Lower semi-continuity for ϕ_2 fails both

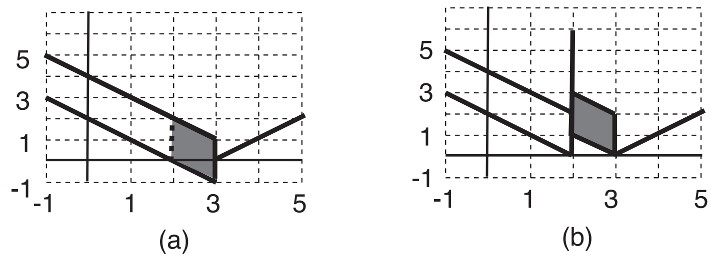

Figure M3.1. Graphs of two correspondences for Exercise 2.

at $x = 2$ and $x = 3$. You can make ϕ_2 lower semi-continuous at $x = 2$ by setting $\phi_2(2) = \{2\}$ and at $x = 3$ by making $\phi_2(3) = \{0\}$.